Week-3 Assignment

Numericals.

Q.2.

'S_1' turns on for 30μs

Time period, $T_s = \frac{1}{10000} = 100\mu s$

$\Rightarrow D = 0.3$

Thus $I_{RMS,S_1} = \sqrt{D} \cdot I_0$

$= 20 \times \sqrt{0.3}$

$I_{RMS,S_1} = 10.95\ A$

Q.3.

'S_1' turns on for 30μs

& turns-off (when S_2 & S_3 is on) for 20μs

$\therefore D = \frac{T_{on}}{T_{on} + T_{off}} = \frac{30}{30+20} = 0.6$

$\Rightarrow I_{RMS,S_1} = 10 \times \sqrt{0.6}$

$I_{RMS,S_1} = 7.74\ A$

Q.5.

$$C = \frac{P_2}{2\pi f_0 \Delta V_0 \cdot V_0} \qquad \text{(from Lec-11)}$$

$$\Rightarrow \quad 2.2 \times 10^{-3} = \frac{3300}{2 \times 3.14 \times 50 \times \Delta V_0 \times 400}$$

$$\Delta V_0 = 11.942\ V$$